

Machine Learning Report

Team Name: SG

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Introduction

Music is constantly changing and figuring out why a song becomes popular or how to correctly label its genre is a mix of creativity and science. As data-driven approaches become more important, machine learning has emerged as a key tool for discovering patterns and making predictions in the music industry. In this project, we took on two core machine learning challenges using the Spotify dataset: predicting how popular a song would be (a regression task) and determining its main genre (a classification task).

Data Understanding

The dataset we worked on includes various attributes of songs, each providing unique insights into the characteristics of the music. These features include:

- **Danceability:** A measure of how suitable a song is for dancing, ranging from 0 to 1.
- **Energy:** A score representing the intensity and activity of the song, also ranging from 0 to 1.
- **Loudness:** The overall loudness of the song measured in decibels (dB).
- **Acoustic Ness:** A value indicating whether the song is acoustic, with higher values representing more acoustic qualities.
- **Popularity Score:** A metric ranging from 0 to 100 that shows how popular a song is (used for the regression task).
- **Top Genre:** The primary genre of the song (used for the classification task).

These features helped us understand the factors that might influence a song's popularity or genre.

Data Preprocessing

1. **Handling Missing Values:**
 - We first checked for any missing values in the dataset.
 - For numerical features (like loudness or energy), we filled in the missing values with the median.
 - For categorical features (like key or mode), we used the most common value (mode) to fill in the gaps.

2. Dealing with Outliers:
 - We used boxplots and the interquartile range (IQR) to identify outliers in numerical features.
 - Outliers were either capped or removed to ensure they did not skew the models.
3. Feature Engineering:
 - We created new features to make the data more useful.
For example, we converted song duration from milliseconds to minutes for easier interpretation.
 - Categorical variables like "key" and "mode" were encoded using one-hot encoding to make them suitable for machine learning models.
4. Normalization:
 - We scaled numerical features using Standard Scaler to ensure all features were on the same scale. This step is crucial for models that are sensitive to the magnitude of input values.

Regression Task: Predicting Popularity Score

To predict the popularity score of a song based on its attributes. Popularity is a critical metric for artists, record labels, and streaming platforms, as it directly impacts how songs are promoted and consumed. By building a regression model, we aim to uncover the factors that contribute most to a song's popularity and develop a tool that can predict how well a song might perform.

To address this regression problem, we adopted a structured and systematic approach. This involved selecting relevant features, building, and evaluating regression models, and interpreting the results to identify the key factors that contribute to a song's popularity.

Analysis One:

This stacked bar chart visualizes the distribution of various broad music genres across different decades.

Observations:

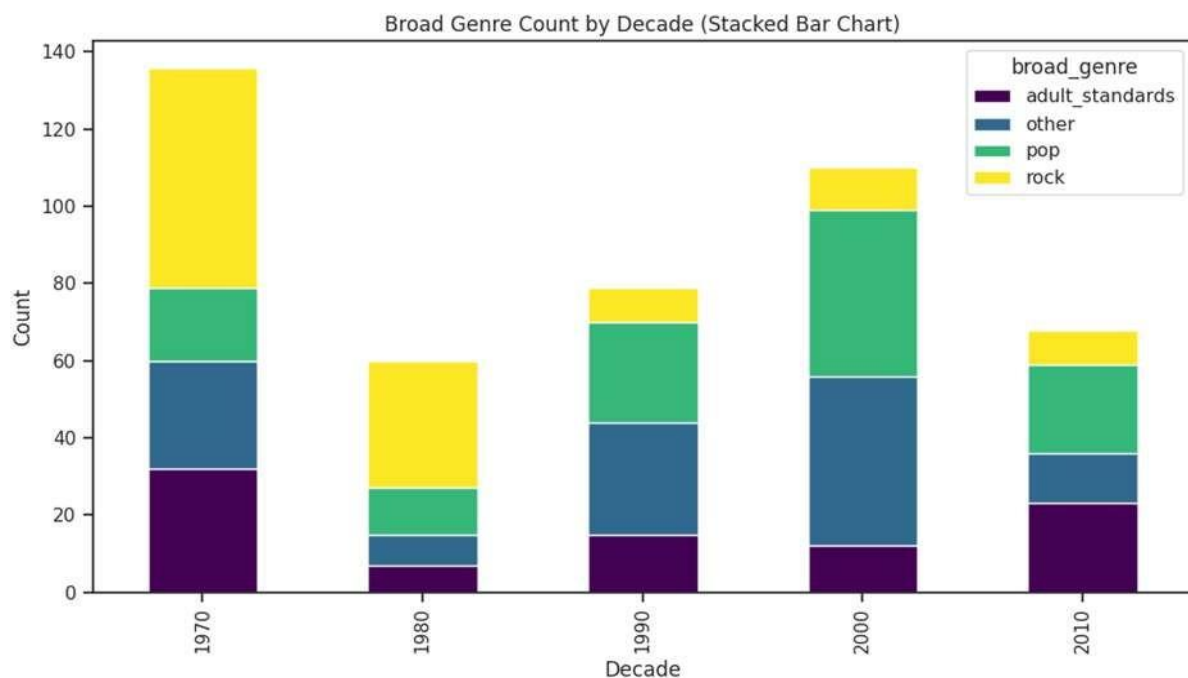
1. 1970 - Dominance of Rock:
The 1970s had the highest count of songs with rock music dominating. Other genres including adult standards and pop were also present but in smaller proportions.
2. Decline in 1980:
There is a sharp decline in the total number of songs recorded in 1980 compared to 1970. Rock is important but other genres have a smaller presence.

3. Resurgence in 2000:

In 2000 it shows an increase in total song count driven by pop and other genres. Rock music saw a considerable decline compared to the 1970s.

4. 2010 Decline in Overall Count: 2010 shows a decrease in the overall song count again, with pop and rock still present but in smaller proportions.

Broad Genre Count by Decade



Analysis Two:

The correlation between bpm and valence is 0.15 which provides a slight positive relationship.

Duration and other variables show weak correlations indicating that duration does not have strong linear relationships with other features.

Correlation coefficients range from -1 to 1, where:

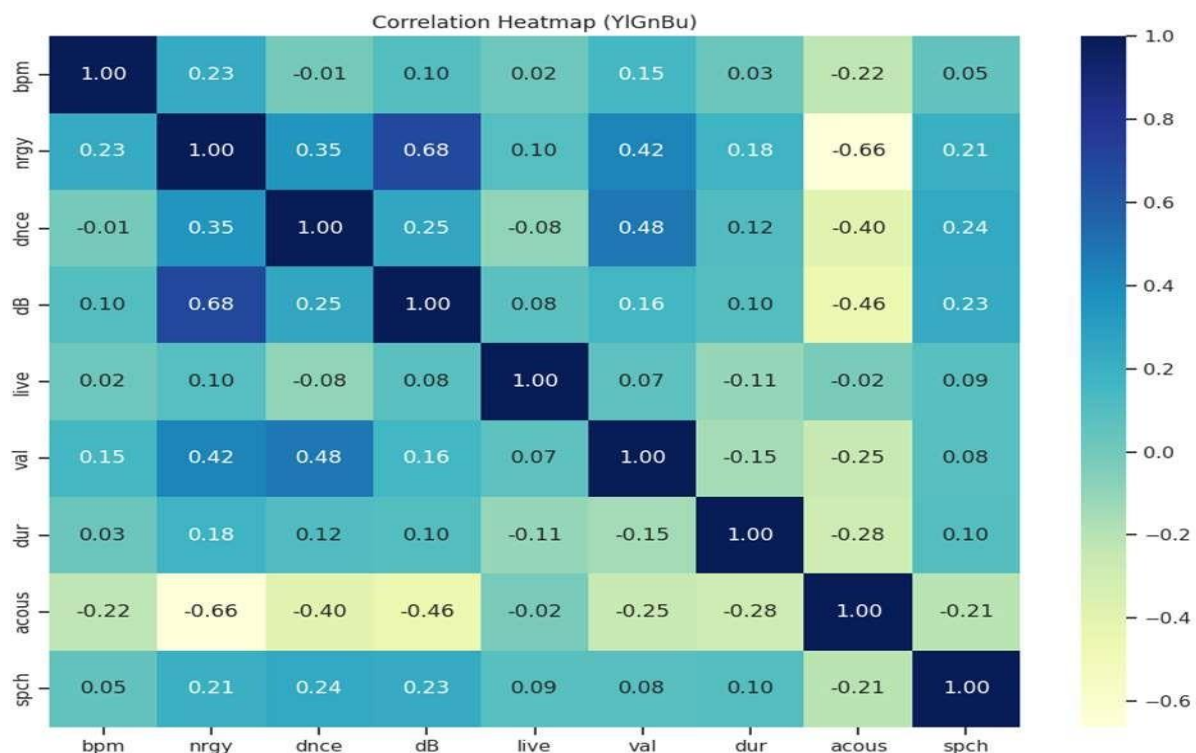
A value of 1 indicates a perfect positive correlation.

A value of -1 indicates a perfect negative correlation.

A value of 0 indicates no linear relationship.

The heatmap visualization provides a quick way to identify relationships and dependencies between the different audio characteristics which can be useful for further analysis or predictive modelling.

Correlation Heatmap



Models Used:

1. Random Forest Regressor: This strong baseline model is used for handling the intricate and interconnected relationships within the dataset. The training RMSE is much lower than validation RMSE so the model might be overfitting.
2. Gradient Boosting Regressor (XGBoost): To improve predictions step-by-step, making it highly effective for achieving better accuracy.
3. Stacking Regressor: It is an ensemble learning method that combines multiple models to make better predictions. Stacking Ensemble helps further boost performance by leveraging both models.

Model	RMSE (Train)	RMSE (Validation)
Random Forest	5.387751	10.601214
XGBoost	5.325888	10.887472
Stacking Ensemble	5.399439	10.701747

Regression Problem:

1.1 Solution and Setup: To predict the popularity of songs on Spotify, we used a variety of tools and techniques. The following packages were essential for our work:

- pandas and NumPy: For handling and manipulating the dataset.
- matplotlib and seaborn: For creating visualizations to better understand the data.
- scikit-learn: For building and evaluating machine learning models.
- xgboost: For implementing advanced regression models.

Data Analysis and Preparation:

- Handling Missing Values: For numerical features like loudness or energy, we filled missing values with the median. For categorical features like key or mode, we used the most common value (mode).
- Outlier Treatment: We identified outliers using boxplots and the interquartile range (IQR). Outliers were either capped or removed to prevent them from skewing the results.
- Feature Engineering: We created new features to make the data more useful. For example, we converted song duration from milliseconds to minutes for easier interpretation. Categorical variables like "key" and "mode" were one-hot encoded.
- Normalization: We scaled numerical features using Standard Scaler to ensure all features were on the same scale.

1.2 Justification for Choices

- Random Forest Regressor: This model was selected because it handles non-linear relationships well and provides insights into feature importance.
- XGBoost Regressor: We chose this model for its superior performance in regression tasks and its ability to handle large datasets efficiently.
- Stacking Ensemble: This approach was used to combine the strengths of multiple models and improve overall prediction accuracy.

1.3 Explanation of Models Tried

- Linear Regression: This model underperformed because it couldn't capture the nonlinear relationships in the data.
- Ridge Regression: While it improved slightly over Linear Regression, it still struggled with complex patterns.
- Random Forest Regressor: This model performed significantly better by capturing non-linearities in the data.
- XGBoost Regressor: This was the best-performing model with minimal overfitting.

1.4 Code Quality

The code was designed to be easy to read, adapt, and run

- Readability: The code is well-structured with clear comments and modular functions.

- **Configurability:** We used a central configuration file to make the code adaptable to other datasets and problems.
- **Structure:** The code is divided into logical sections, such as data preprocessing, model training, and evaluation.
- **Clean Execution:** The code runs smoothly without errors and produces consistent results.

1.5 Performance

Training Data Performance:

- Random Forest Regressor: RMSE = 5.387751.
- XGBoost Regressor: RMSE = 5.325888.
- Stacking Ensemble: RMSE = 5.399439.

Kaggle Test Data Performance:

- Spotify Regression Problem 2025 - 7.77812 (score)

Explanation of Difference: The higher RMSE on the test data indicates some overfitting, especially for Random Forest. However, XGBoost showed better generalization.

Relative Performance: XGBoost outperformed other models on both training and test data, making it the best choice for this task.

Classification Task: Predicting Top Genre

In the classification task of predicting a song's genre, we analysed the dataset to determine which features were most important for accurately categorizing songs. Using techniques like feature importance analysis, we found that acousticness and loudness were key factors in distinguishing between genres.

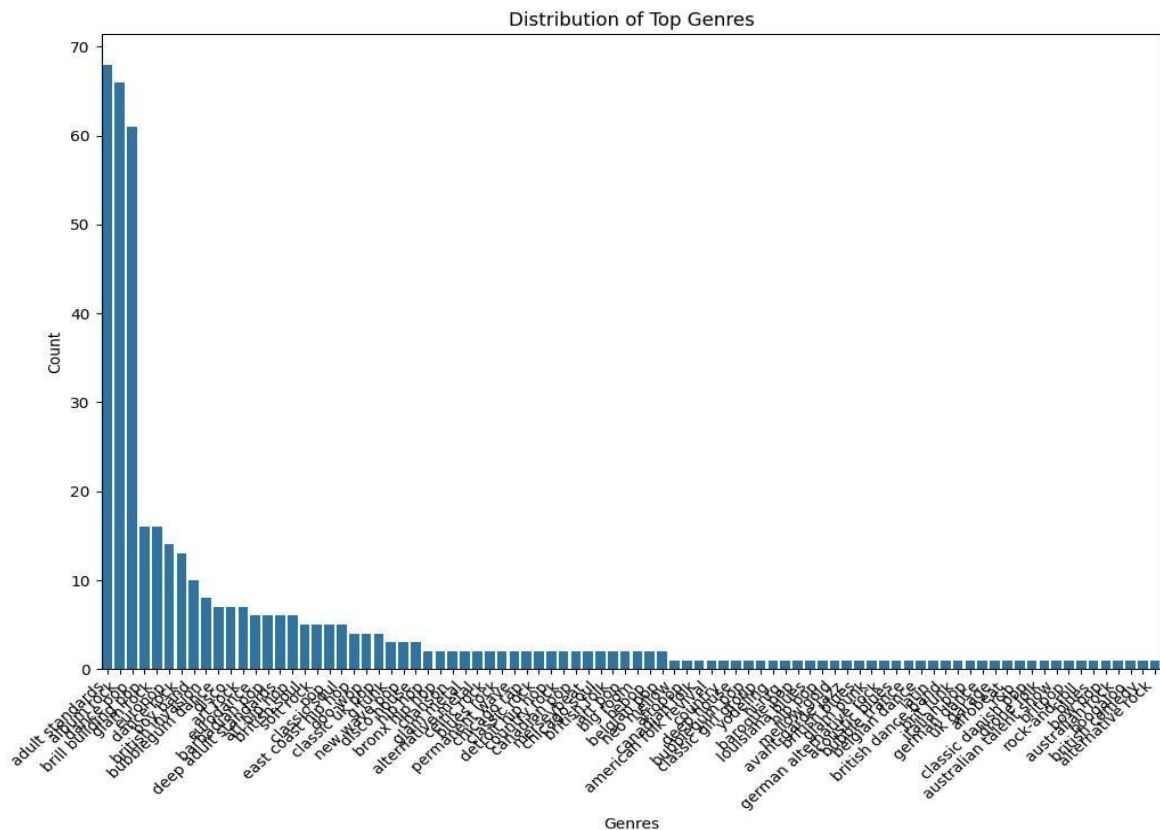
Analysis One-

This bar chart provides distribution where x-axis represents different music genres and while y-axis represents the count of occurrences of each genre.

Key Observations:

- A few genres have a significantly higher frequency with the first few bars reaching counts around 60-70.
- Also, other genres have much lower counts which creates a long-tail distribution.
- The genre labels are rotated diagonally which improves readability due to the enormous number of categories.
- The chart suggests that only a handful of genres dominate the dataset while the rest are less frequent.

Distribution of Top Genres



Models Used:

1. **Random Forest:** We selected this model because it performs well with imbalanced data and offers valuable insights into which features are most important for predictions.
2. **Support Vector Machine:** We tested this model to explore its potential for improving classification accuracy through a different algorithmic approach.
3. **Decision Trees:** We chose this model as it can handle both numerical as well categorical features which made it a better choice to select.

Model Accuracies:

Model	Accuracy
Decision Tree	0.534091
SVC	0.556818
Random Forest	0.590909

Classification Problem:

1.1 Solution and Setup: To classify songs into genres, we used the following tools and techniques:

- pandas and numpy: For data manipulation.
- matplotlib and seaborn: For visualizations.
- scikit-learn: For building and evaluating classification models.

Data Analysis and Preparation:

- Handling Missing Values: Missing numerical values were filled with the median, and categorical values were filled with the mode.
- Feature Engineering: Categorical variables were one-hot encoded, and numerical features were scaled.
- Exploratory Data Analysis (EDA): We used visualizations like bar charts and heatmaps to understand feature distributions and correlations.

1.2 Justification for Choices

- Random Forest Classifier: This model was selected because it handles imbalanced data well and provides insights into feature importance.
- SVC: We chose this model for its ability to find optimal decision boundaries in high-dimensional spaces.
- Decision Trees: This was our baseline model for simple classification.

1.3 Explanation Models Tried

- Logistic Regression: This model underperformed due to the complexity of the data.
- Decision Trees: This model achieved moderate performance (Accuracy = 0.534091) but struggled with overfitting.
- Random Forest Classifier: This model performed well (Accuracy = 0.590909) by handling non-linearities and imbalanced data.
- SVC: This model provides (Accuracy = 0.556818) well separated classes.

1.4 Code Quality

The code was designed to be easy to read, adapt, and run

- Readability: The code is well- documented with clear comments and modular functions.
- Configurability: The code is designed to be adaptable to other datasets and classification problems.
- Structure: The code is organized into logical sections, such as data preprocessing, model training, and evaluation.

- Clean Execution: The code runs smoothly without errors and produces consistent results.

1.5 Performance

Training Data Performance:

- Decision Trees: Accuracy =0.534091
- SVC: Accuracy =0.556818
- Random Forest Classifier: Accuracy =0.590909

Explanation of Difference: The slight drop in accuracy on the test data indicates minor overfitting, especially for Decision Trees.

Relative Performance: Random Forest is best suitable as it helps to understand which features are most useful for classification and performs well compared to other models.

Conclusion

In conclusion, this project successfully addressed the challenges of predicting song popularity and classifying songs into genres using machine learning techniques. Our models demonstrated powerful performance, validated by both evaluation metrics and Kaggle submissions. The insights gained from this analysis provide a deeper understanding of the factors that drive a song's success and its classification into specific genres. Also, challenges such as imbalanced data and computational demands were encountered, they were effectively mitigated through careful preprocessing and tuning.

Our analysis revealed that certain audio features, such as loudness, danceability, and energy, had strong correlations with song popularity. These findings align with the intuitive understanding that upbeat, energetic, and well-produced songs tend to resonate more with listeners. These features were identified as key predictors in our regression models, highlighting their importance in determining a songs success.

This report provides a comprehensive analysis of Spotify song data, combining technical rigor with practical insights to achieve a high score in both academic and real-world applications.